

Quiz 1 – Calculus I (Class 64)

1. Solve the inequality $3 \leq |2x - 5| \leq 7$. Express the solution as a *set* notation and an *interval* notation.
2. Find the coordinate (a, b) if a line passes through $(-2, 0)$ and (a, b) has gradient 2, and a line passes through $(3, 5)$ and (a, b) has gradient $3/4$.
3. Given two functions $f(x) = \sqrt{x^2 - 9}$ and $g(x) = \frac{1}{x}$.
 - (a) Find $\left(\frac{f}{g}\right)(x)$ and its domain.
 - (b) Find $(f \circ g)(x)$ and its domain.
4. Given $f(x) = 2 + \sqrt{25 - x^2}$ with $0 \leq x \leq 5$.
 - (a) Find $f^{-1}(x)$ and its domain.
 - (b) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ in one coordinate system.

1. We split the compound absolute-value inequality $3 \leq |2x - 5| \leq 7$ into two cases.

(a) $2x - 5 \geq 0$ (i.e. $x \geq 5/2$):

$$3 \leq 2x - 5 \leq 7 \implies 8 \leq 2x \leq 12 \implies 4 \leq x \leq 6.$$

(b) $2x - 5 < 0$ (i.e. $x < 5/2$):

The inequality becomes $3 \leq -(2x - 5) \leq 7$, i.e. $3 \leq 5 - 2x \leq 7$:

$$3 \leq 5 - 2x \leq 7 \implies -2 \leq -2x \leq 2 \implies -1 \leq x \leq 1.$$

Taking the union of both cases:

$$\boxed{S = \{x \in \mathbb{R} \mid -1 \leq x \leq 1 \text{ or } 4 \leq x \leq 6\} = [-1, 1] \cup [4, 6].}$$

2. Let (a, b) be the unknown point.

From the gradient conditions:

(a) Line through $(-2, 0)$ and (a, b) has gradient 2:

$$\frac{b - 0}{a - (-2)} = 2 \implies b = 2a + 4. \quad (1)$$

(b) Line through $(3, 5)$ and (a, b) has gradient $3/4$:

$$\frac{b - 5}{a - 3} = \frac{3}{4} \implies 4b = 3a + 11. \quad (2)$$

Substituting (1) into (2):

$$4(2a + 4) = 3a + 11 \implies 8a + 16 = 3a + 11 \implies a = -1, \quad b = 2.$$

$$\boxed{(a, b) = (-1, 2).}$$

Verification:

- Gradient $(-2, 0) \rightarrow (-1, 2)$: $\frac{2}{1} = 2$. ✓
- Gradient $(3, 5) \rightarrow (-1, 2)$: $\frac{-3}{-4} = \frac{3}{4}$. ✓

3. Given $f(x) = \sqrt{x^2 - 9}$ and $g(x) = \frac{1}{x}$.

Preliminary domains:

- $D_f = (-\infty, -3] \cup [3, +\infty)$ (need $x^2 - 9 \geq 0$).
- $D_g = \mathbb{R} \setminus \{0\}$ (need $x \neq 0$).

(a) $\left(\frac{f}{g}\right)(x)$:

$$\left(\frac{f}{g}\right)(x) = \frac{\sqrt{x^2 - 9}}{\frac{1}{x}} = x\sqrt{x^2 - 9}.$$

Since $x \leq -3$ or $x \geq 3$ already excludes $x = 0$:

$$\boxed{\left(\frac{f}{g}\right)(x) = x\sqrt{x^2 - 9}, \quad D_{f/g} = (-\infty, -3] \cup [3, +\infty).$$

(b) $(f \circ g)(x)$:

$$(f \circ g)(x) = f\left(\frac{1}{x}\right) = \sqrt{\frac{1}{x^2} - 9} = \frac{\sqrt{1 - 9x^2}}{|x|}.$$

Domain requires $x \neq 0$ and $g(x) \in D_f$, i.e. $\frac{1}{x} \leq -3$ or $\frac{1}{x} \geq 3$:

- $\frac{1}{x} \geq 3$: $\frac{1 - 3x}{x} \geq 0 \Rightarrow 0 < x \leq \frac{1}{3}$.
- $\frac{1}{x} \leq -3$: $\frac{1 + 3x}{x} \leq 0 \Rightarrow -\frac{1}{3} \leq x < 0$.

$$\boxed{(f \circ g)(x) = \frac{\sqrt{1 - 9x^2}}{|x|}, \quad D_{f \circ g} = \left[-\frac{1}{3}, 0\right) \cup \left(0, \frac{1}{3}\right].$$

4. Given $f(x) = 2 + \sqrt{25 - x^2}$, $0 \leq x \leq 5$.

(a) Let $y = 2 + \sqrt{25 - x^2}$. Solving for x :

$$\begin{aligned} y - 2 &= \sqrt{25 - x^2}, \\ (y - 2)^2 &= 25 - x^2, \\ x^2 &= 25 - (y - 2)^2, \\ x &= \sqrt{25 - (y - 2)^2} \quad (\text{positive root since } x \geq 0). \end{aligned}$$

Swapping $x \leftrightarrow y$:

$$\boxed{f^{-1}(x) = \sqrt{25 - (x - 2)^2}.$$

Domain of f^{-1} = Range of f . Since $f(0) = 7$, $f(5) = 2$, and f is strictly decreasing: $D_{f^{-1}} = [2, 7]$.

(b) Both curves are quarter-circle arcs (radius 5) symmetric about the line $y = x$.

